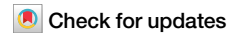


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Evidence for common fungal networks among plants formed by a Dark Septate Endophyte in *Sorghum bicolor*



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Fungal connections among plants, popularly known as the “wood wide web,” captured the interest of scientific and public audiences because these connections may facilitate increased growth, improved survival, nutrient transfer, and communication among plants. Research on these fungal networks has focused almost exclusively on known plant symbionts called mycorrhizal fungi. However, many non-mycorrhizal fungi also form ecologically important associations with plants. If non-mycorrhizal fungi such as Dark Septate Endophytes (DSEs) can form common networks among plants, then fungal connections among plants are likely more complex and prevalent than previously thought. In this study, we ask whether a common DSE can form hyphal connections between plants, improve plant biomass, and move water between them. Using a lab system with donor and receiver plants, we find that DSE hyphae crossed air gaps to physically connect plants. Receiver plants that were connected to a fungal network had higher biomass than those that were not. A water-soluble dye injected into donor plant leaves was detected in receiver leaves, but only when plants were connected via the fungal network. These results provide the first lab-based evidence that common non-mycorrhizal networks can occur and suggest that fungal networks among plants may extend beyond mycorrhizal fungi.

The widespread associations between plants and fungi are gaining renewed scientific and public attention, particularly with growing interest in the phenomenon popularly known as the “wood wide web”^{1–3}. Traditionally studied as Common Mycorrhizal Networks (CMNs), these interactions involve mycorrhizal fungi forming hyphal connections among plant roots³. CMNs are thought to have diverse ecological consequences, including enhanced access to beneficial microbes, transfer of carbon and nutrients among plants, interplant signaling that may reduce herbivory, and increased plant growth^{1,4–6}. This growing interest has prompted reexamination of existing evidence^{1,7,8}. One emerging aspect of reexamination of CMNs is that fungi beyond those forming classical mycorrhizal associations may also connect spatially separated plants^{9,10}. We define mycorrhizae (myco = fungus, rhiza = root) as the intimate symbioses formed between the roots of over 90% of plants and certain groups of fungi, where there is a bidirectional exchange of resources^{11,12}. Each type of mycorrhizal association produces characteristic and distinct fungal structures, such as the Hartig net, arbuscules, and coils. Under this definition, groups of mycorrhizal fungi include ectomycorrhizal, arbuscular mycorrhizal, monotropoid mycorrhizal, orchid mycorrhizal, ericoid mycorrhizal, and Mucoromycotina fine root

endophytes^{12–14}. Along with this diversity of mycorrhizal fungi, there is also tremendous diversity of non-mycorrhizal plant-associated fungi that may form common networks among plants⁹.

One group of non-mycorrhizal fungi that may form common networks among plants is the dark septate endophyte (DSE), a polyphyletic group of root-associated fungi that are defined by melanized hyphae with regular walls between cells (septa)^{15,16}. DSEs have been observed in the roots of both woody and herbaceous plants in diverse ecosystems, including tundra, agricultural fields, deserts, and forests, where DSEs strongly influence the species composition and diversity of other root and soil microbes¹⁷. Similar to mycorrhizal fungi and sometimes even classified as mycorrhizal, DSE interactions with plants span a mutualism-parasitism continuum, where their effects on plants range from positive to negative depending on environmental context^{17–21}. Under conditions conducive to mutualisms, plants may provide carbon to the DSEs and, in turn, DSEs enhance plant growth and uptake of soil resources, including nitrogen and phosphorus^{18,20}. A recent study estimated that a similar amount of net primary productivity was allocated to DSEs (4.6%) as to other groups of plant-associated fungi, including mycorrhizal fungi²². However, a bidirectional exchange of resources has not been

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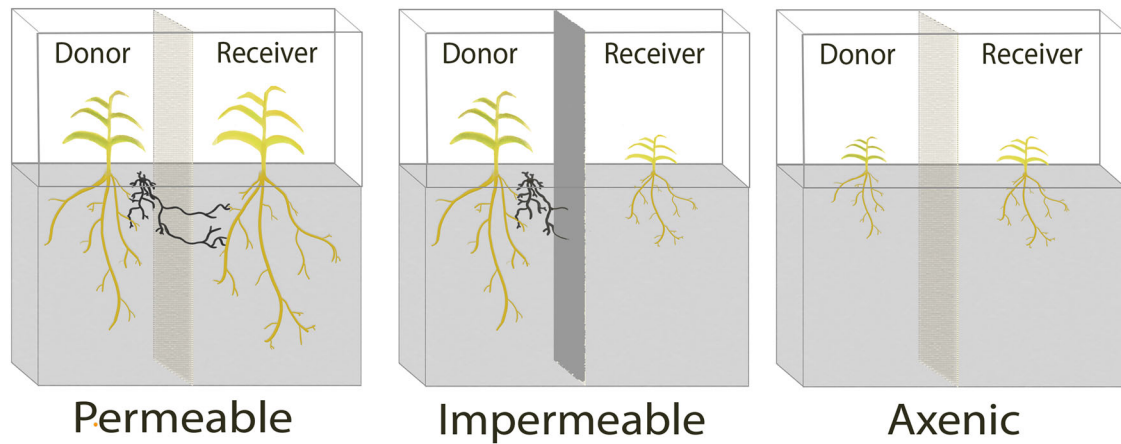
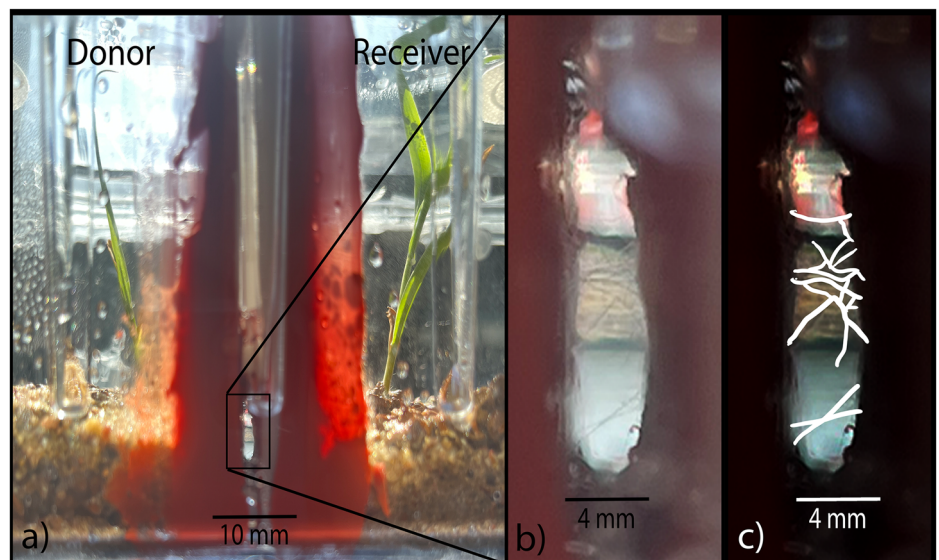


Fig. 1 | Conceptual diagram of the three experimental treatments. In the Permeable treatment, a permeable barrier separates donor and receiver plants; the dark septate endophyte (DSE) is added to the donor chamber and can cross the barrier to colonize the receiver plant. In the Impermeable treatment, an impermeable barrier separates the chambers; the DSE is added to the donor side but cannot

reach the receiver plant. In the Axenic treatment (no culturable microbes), a permeable barrier is used, but the DSE is not introduced, so neither plant associates with a DSE. See Supplementary Fig. 1 for barrier design details. Box dimensions: 24.5 × 9.7 × 8.2 cm (length × width × height).

Fig. 2 | Side view of the experimental box in the Permeable treatment. A top view of the box is provided in Supplementary Fig. 3. In **a**, the donor plant is on the left and the receiver plant is on the right. The chambers are separated by a permeable barrier with a 3 mm air gap (no media present inside the air gap), across which DSE hyphae are visible in the inset (**b**). Panel **c** is an annotated version of Panel **b** highlighting the hyphal structures connecting the donor and receiver chambers with overlay lines. See Fig. 1 for a conceptual diagram of these boxes. Scale bars represent 10 mm for **a** and 4 mm for **b** and **c**.



demonstrated in DSE-plant interactions, nor have specific exchange structures or symbiotic signaling pathways been documented^{16,21}. Despite the lack of proof for a nutritional symbiosis between DSEs and plants, DSEs are an ideal group to study for the potential of common non-mycorrhizal networks because they colonize both mycorrhizal and non-mycorrhizal plants, exhibit low host specificity and possess strong saprobic abilities that may facilitate soil nutrient exchange with plants^{15,16}.

We conducted a laboratory experiment (Fig. 1) to test three hypotheses: (H1) a DSE can form a Common Fungal Network connecting two spatially separated plants; (H2) plant biomass is higher when plants have access to such a fungal network; and (H3) materials can move between plants connected by a Common Fungal Network, assessed via a tracer dye injected into donor plants. To test these hypotheses, we established three experimental treatments (Fig. 1; Supplementary Table 1): 1) “Permeable”: donor and receiver plants are separated by a permeable barrier (Supplementary Fig. 1); the DSE is added to the donor chamber and can cross the barrier to interact with the receiver plant. 2) “Impermeable”: an impermeable barrier separates donor and receiver plants; the DSE is added to the donor chamber but cannot reach the receiver. 3) “Axenic”: donor and receiver plants are separated by a

permeable barrier, but the DSE is not introduced, leaving both donor and receiver plants axenic (i.e., free of culturable microbes). We used an agriculturally important crop, *Sorghum bicolor* (L. Moench), as the model plant and a common DSE, *Alternaria alternata* (Nees), as the model fungus. To address concerns about evidence for fungal networks being confounded with soil diffusion pathways¹, a 3-mm wide air gap separated the donor and receiver plants in this experiment, and there was no evidence of dye movement through condensation or any means other than through the DSE (Fig. 2, Supplementary Fig. 1). In addition to our “Main” experiment, we conducted a “Follow-Up” experiment with accessory treatments to further verify that the results observed were due to connection by the DSE and not other variables (Supplementary Table 1).

Results and Discussion

Physical connections among plants through DSE

Our results provide preliminary evidence that a non-mycorrhizal fungus may connect plants in common networks, increasing the biomass of the networked plants and moving water between plants. Moreover, we found evidence in support of H1 with hyphae from donor plants crossing an air

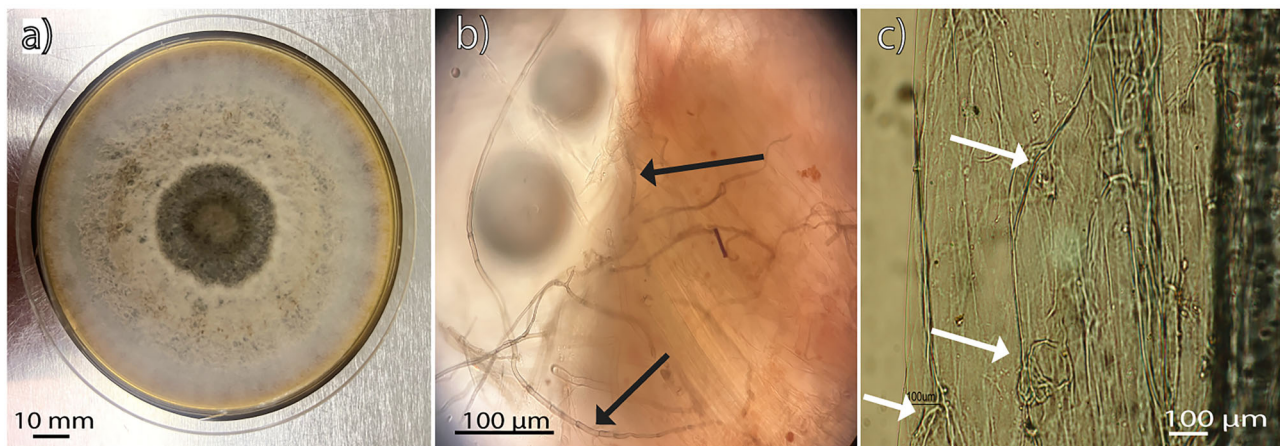


Fig. 3 | Morphology and Root Colonization of *Alternaria alternata* in Culture and Plant Tissues. **a** Top view of the DSE culture (*Alternaria alternata*) used in this experiment on Potato Dextrose Agar. Scale bar represents 10 mm. **b** DSE hyphae outside of and going into roots of a receiver plant in the Permeable treatment, stained

with acid fuchsin (pink). **c** Internal DSE colonization of a sorghum root, cleared with KOH and stained with Trypan Blue. In **b** and **c**, arrows point to DSE hyphae and scale bars represent 100 µm.

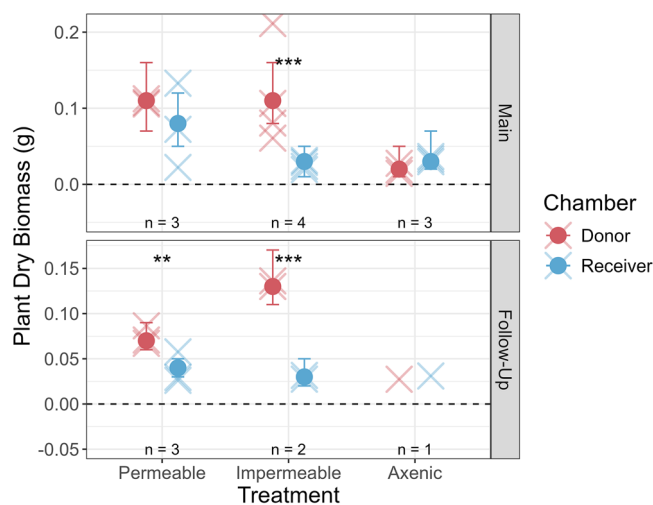


Fig. 4 | Plant biomass across treatments and experimental rounds. Filled circles with error bars represent estimated marginal means of total plant dry biomass (g) per plant ($\pm 95\%$ confidence intervals), derived from generalized linear models (quasipoisson, log link). Estimates are shown for each chamber type (Donor = red, Receiver = blue), treatment (Permeable, Impermeable, Axenic), and experimental round (Main, Follow-Up). Semi-transparent x-marks show individual plant biomass values. Significance of pairwise contrasts between chambers within each treatment is indicated (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$), based on back-transformed comparisons. Sample sizes (number of paired donor-receiver samples) are shown below each treatment group. Treatments with fewer than two replicates were excluded from statistical analyses and are shown only as individual points without summary statistics.

gap and colonizing the roots of receiver plants in the Permeable treatment from the main experiment but in none of the Impermeable or Axenic treatments (Figs. 2, 3; Supplementary Table 2). The DSE was also observed directly connecting the roots of plants grown in a clear substrate for demonstration purposes (Supplementary Fig. 2). These findings represent an initial step toward exploring the potential for network formation among non-mycorrhizal fungi and raise the possibility that CMNs are just one of several types of microbial networks connecting plants.

Biomass effects of access to a Common Fungal Network

In support of H2, plant biomass was higher in sorghum receiver plants that had access to the DSE compared to those that did not (Fig. 4; Supplementary Fig. 3; Supplementary Table 3; Supplementary Table 4). Receiver plants in

the Impermeable treatment (with no access to the DSE) had significantly lower biomass than donor plants ($p < 0.001$), while no significant differences were observed between donor and receiver biomasses in the Permeable ($p = 0.28$) and Axenic ($p = 0.43$) treatments of the Main experiment (Fig. 4; Supplementary Table 3; Supplementary Table 4). These results are in line with one of the hypothesized benefits of CMNs to plants, which is that access to a fungal network provides a ready source of fungal inoculum to the receiver plant⁴. This hypothesized benefit of networks among plants does not require unbroken hyphal connections, transfer of resources among plants, or even exclusion of a soil diffusion pathway. Given how poorly plants grew in this experiment without access to the DSE (Fig. 4; Supplementary Table 4), the observed increases in biomass associated with the DSE network suggest a potential benefit of Common Fungal Networks to plants. Notably, we observed variability in plant responses, with some receiver plants showing no biomass benefit despite network connections, especially in the Follow-Up round of experimentation. This variability suggests context-dependent effects of DSE-mediated networks, aligning with findings from previous CMN studies, but were not fully explored in this experiment^{1,2,3,24}.

Interplant water transfer through DSEs

In support of H3, the water-soluble dye acid fuchsin, injected into all donor plant leaves, was detected in receiver plant leaves only in the Permeable treatment, where donor and receiver plants were connected solely through the DSE network (Fig. 5; Supplementary Table 5; Supplementary Table 6). Receiver plants were never injected. The dye served as a tracer for water movement between plants, and dye content was quantified based on peak absorbance values and dry leaf mass (Supplementary Fig. 4). No evidence of dye leakage was observed around plants or in the substrate. Dye was consistently detected in donor plant leaves in both the Permeable and Impermeable treatments (Fig. 5), supporting the reliability of this method. In contrast, donor plants in the Axenic treatment did not show detectable dye levels, likely due to poor growth in the absence of fungal association, which may have limited their capacity to take up and translocate the dye.

Donor and receiver plant tissues appeared visibly pink in the Permeable treatment (Fig. 3), and there was no significant difference in dye content between donor and receiver plants in this treatment of the Main experiment ($p = 0.57$). In contrast, receiver plants in the Impermeable treatment contained significantly less dye than their paired donor plants ($p < 0.01$; Fig. 5; Supplementary Table 5). In both the Main and Follow-Up experiments, the 95% confidence intervals for dye content in receiver plants in the Impermeable treatment and all plants in the Axenic treatment included zero, while they were positive for both donor and receiver plants in the

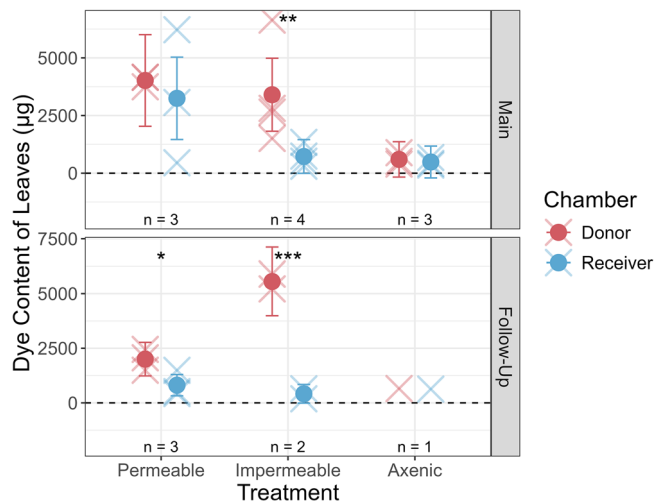


Fig. 5 | Dye accumulation in shoot tissue across treatments and experimental rounds. Filled circles with error bars represent estimated marginal means of dye content in plant leaves for each treatment ($\pm 95\%$ confidence intervals), derived from generalized linear models (quasipoisson, identity link). Significance of pairwise contrasts is indicated (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Semi-transparent x-marks show predicted dye content (μg) in dried leaf material for individual plants, calculated using a linear model relating absorbance at 545 nm to known dye concentrations. Sample sizes are indicated below each group, where one sample is composed of one donor plant and one receiver plant. Treatments with fewer than two replicates were excluded from statistical analyses and are shown only as individual data points.

Permeable treatment and for donor plants in the Impermeable treatment (Supplementary Table 6). These results support H3 by suggesting water movement from donor to receiver plants via a common DSE network. They also support H1, indicating that a common DSE network connected the plants: the movement of dyed water between chambers could not have occurred without a mycelial pathway bridging the air gap between the donor and receiver chambers. As with the biomass data, we observed variability in dye movement both within treatments and across experimental rounds. This likely reflects context-dependent dynamics not captured in our design but documented in the CMN literature^{1,23,24}. Nonetheless, the presence of positive cases of dye transfer in this experiment provides preliminary support for the concept of DSE-mediated connections between plants.

Toward a broader framework for Common Fungal Networks

To the authors' knowledge, these results represent the first experimental indication that common DSE networks may occur. Given the prevalence of DSEs and other non-mycorrhizal plant-associated fungi, this finding has implications for how we understand the potential connectivity of plants in ecosystems. DSEs are found nearly everywhere they are looked for, across diverse habitats, both outside of and in association with many different plant lineages, with low known host specificity¹⁵. DSEs are found both in mycorrhizal plant lineages (e.g., Pinaceae¹⁵) and non-mycorrhizal plant lineages (e.g., Brassicaceae²⁵), meaning that it is possible for mycorrhizal and non-mycorrhizal networks to be acting independently, synergistically, and antagonistically. While CMNs remain an important phenomenon to study, their non-mycorrhizal counterparts should also be considered as potential agents for movement of nutrients, water, and signaling compounds in future research. Additionally, the presence of common non-mycorrhizal networks creates support for using the broader term "Common Fungal Network" to describe the more general fungal connections that occur among plants⁹, of which CMNs are just one type. This framework for understanding fungal connections among plants creates grounds for many exciting areas of study, including the materials transferred in different types of Common Fungal Networks, synergism and antagonism among fungal networks, and the spatial prevalence of each type.

The finding of increased biomass in networked plants in this experiment is especially notable given that the DSE genus used in this experiment, *Alternaria*, contains many species often considered plant pathogens^{26,27}. Other DSE taxa should be examined in future experiments, and considering the potential value in finding fungal inoculum sources to improve crop biomass²⁸, this experiment supports the possibility that DSEs often considered pathogenic may not always act as such. Much of the CMN research has focused on woody plants in forests, but Common Fungal Networks may be important for non-woody plants (such as the crop plant *Sorghum bicolor* in this experiment) and other plants in agricultural settings where DSEs that improve crop biomass could readily form associations with multiple plants.

Although the mechanisms underlying nutritional exchange between DSEs and host plants remain unclear, several plausible pathways have been proposed. Carbon flow from plants to DSEs has been documented in a few cases^{29,30}, and root exudates may serve as a readily available carbon source, particularly when carbon is not limiting for the plant. This carbon benefit to DSEs may arise passively as a by-product of root activity¹⁶. In turn, plants may benefit from DSE associations through enhanced nitrogen acquisition³¹. These benefits could be mediated by DSE-driven nutrient mobilization in the surrounding soil or by hormonal signaling that alters plant nutrient uptake or growth¹⁶. While the exact mechanisms of DSE-plant associations remain unresolved, the range of potential interactions offers rich opportunities for future research.

While highly controlled laboratory experiments are well-suited for testing the potential existence of common non-mycorrhizal networks, they are limited in their ability to assess the prevalence and ecological significance of these networks under field conditions. Furthermore, our study had relatively low replication and an exploratory design, which restricts the strength of the conclusions we can draw. Thus, the results should be interpreted as preliminary evidence and observed trends rather than definitive proof of DSE-mediated plant connectivity. Future research with larger sample sizes, diverse fungal taxa, and field-based experiments is necessary to verify and extend these findings and clarify the ecological roles of common DSE networks. Prior field studies in arid ecosystems have documented plant connectivity via biocrusts, which include DSEs among their constituents^{32–35}. These systems, involving multiple organisms that jointly mediate plant interactions, offer a promising context for isolating and evaluating the specific role of DSEs in facilitating belowground networks. Given the widespread occurrence of both mycorrhizal and non-mycorrhizal plant-fungal associations, it is plausible that Common Fungal Networks, regardless of fungal guild, may be a widespread feature of plant ecosystems.

Methods

Experimental system design

We created experimental boxes by modifying the "We-V Box" with dual ventilation ports from Magenta LLC (<https://wevitro.magentallc.com/vessel/>; dimensions: 24.5 × 9.7 × 8.2 cm, length × width × height; Supplementary Fig. 3). The boxes were modified by gluing permeable or impermeable barriers in the middle to create two separate chambers in each experimental box (Fig. 1; Supplementary Fig. 1). The Permeable barrier was inspired by Warren et al. (2008)³⁶ and consisted of two pieces of hard autoclavable plastic (Delrin acetal copolymer) glued together with a hole (~6 cm³) cut in them 2 cm above the box floor to allow resource transfer through mesh and a 3-mm wide air gap. Impermeable barriers did not have a hole cut into the Delrin. The mesh consisted of a coarse layer (100-micron diameter) and a fine layer (38-micron diameter), proximal and distal to each chamber, respectively, on both sides of the 3-mm air gap. The coarse mesh protected the fine mesh, and the fine mesh was a barrier to plant roots but allowed fungal hyphae to cross. At harvest, there was no evidence of root penetration through the meshes. The 3-mm air gap in the middle of the barrier prevented diffusion between the two chambers, which allowed us to isolate the movement of materials through the fungal hyphal pathway versus the soil diffusion pathway. The air gap was visible throughout the experiment from the outside of the boxes (e.g., Fig. 2). We did not observe condensation within the air gap, nor the presence of dye on the mesh. Moreover,

if the air gap did not prevent mass flow between chambers, we would have expected to see dye move from donor to receiver chamber in the Axenic treatment, which was not the case (Fig. 5; Supplementary Fig. 5).

We used high-temperature resistant silicone sealant (American Sealants, Inc.) for all gluing and allowed the sealant to cure for at least 24 hours before installing the barriers into the boxes. This sealant can be seen in Fig. 2 and Supplementary Fig. 3 as the red adhesive sealing the barriers in the middle of the boxes. The chambers on either side of the barriers were designated as the donor chamber and receiver chamber of each experimental box.

A mixture of rinsed and dried turf and sand (100 g of MVP Turf from Turf Athletic and 50 g of Quickrete Premium Play Sand) was placed in each chamber and then autoclaved to sterilize the entire experimental box. The boxes were left to sit for at least 48 hours after autoclaving to permit offgassing and ensure the substrate was at room temperature before adding plants and fungi to the boxes.

Experimental procedure

The *Sorghum bicolor* seeds used were the 9300 Grain Sorghum variety obtained from Hancock Seed Company (<https://hancockseed.com/products/9300-grain-sorghum-seed>). Seeds of *S. bicolor* were sterilized by rinsing in 95% ethanol for 2 minutes, 3.75% sodium hypochlorite solution for 20 minutes, then sterile water twice for one minute each. These seeds were germinated in Petri dishes containing 1% Phytagel in the same plant growth tent used for the experiment, with 500 lumens of light intensity on a 15-hour light/9-hour dark schedule and temperatures ranging from 22 °C to 32 °C. Only seeds that showed no microbial growth on the Phytagel medium, indicating the seeds were successfully rendered axenic, were used in the experiment. Forty-eight hours after plating, one sterile germinant was transferred into each chamber of each experimental box. On the same day, 5 mL of filter-sterilized Hoagland nutrient solution and 75 mL of sterile water was added to each chamber^{37,38}. No additional fertilizer or water was added to the boxes during the Main experiment. Experimental boxes were sealed with Parafilm and placed into the previously described plant growth tent. In the Follow-Up experiment, plants began to show signs of stress (leaf yellowing and wilting) early in the trial; therefore, an additional 3 mL of Hoagland's solution and 30 mL of sterile water were added to each chamber two weeks after the experiment began.

Five days after transferring germinants to boxes, germinants that died after transplanting were replaced with germinants that were sterilized and plated at the same time as the ones used initially. This occurred with 9 plants across the two rounds of experimentation, with no trends noticeable across treatment types. After transplant replacement, we added DSE inoculum to the donor chamber of the Permeable and Impermeable treatment boxes. Inoculum of the DSE *Alternaria alternata* was prepared on Potato Dextrose Agar (PDA) in 100-mm diameter Petri dishes filled with 25 mL of media. After 7 days of growth on PDA, an 8-mm diameter cork borer was used to cut uniform inoculum plugs from the growing edge of the fungus, creating inoculum plugs of about 100.5 mm³ volume. For the Axenic controls, we added a sterile plug of PDA of the same size to the donor chamber. These plugs were dropped onto the surface of the growth medium 1 cm next to the seedling in the donor chambers, in the direction closest to the barrier in the middle of the box. The fungus used in this experiment was originally isolated from sorghum seeds. DNA was extracted, amplified using the ITS1F/ITS4 primer pair following Lamit et al. (2014)³⁹, and sequenced via Sanger sequencing, which allowed us to identify the fungus as *Alternaria alternata* (GenBank Accession Number PQ284877.1). We confirmed with compound light microscopy that the hyphae of this fungus are darkly pigmented and have regular septa (see panel b of Fig. 3). In other experiments in our lab, we have found that this isolate can utilize both organic and inorganic forms of nitrogen⁴⁰ and improves the germination of *S. bicolor* seeds.

We utilized three chamber types to control for variables that often confound CMN research, with the treatments summarized in Supplementary Table 1. The Permeable treatment was the experimental box as described above, where the DSE was added to the donor chamber and able to cross the mesh and

air gap and move from the donor chamber to the receiver chamber. The Impermeable barrier treatment had no window in the plastic barrier between chambers, to test whether the DSE and dyed water could move between the chambers through any way other than the window in the plastic barriers. The Axenic treatment had the same setup as the Permeable treatment, but with a plug of sterile PDA instead of a DSE-inoculated plug of PDA.

Every seven days after starting the experiment, we randomly changed the placement of boxes within the growth tent. The fungus typically crossed the barriers in Permeable treatment boxes within two weeks, and the tracer-dye solution was added after four weeks to allow sufficient time for DSE colonization and growth. The tracer-dye solution, containing 0.2 mg/mL acid fuchsin and 0.0018 g/mL ammonium sulfate dissolved in RO water, was autoclaved, and 0.1 mL of this solution was injected into the leaves of the donor plant in each box using a 1 mL BD SafetyGlide Insulin syringe. Receiver plants were never injected in any treatment. While this injection resulted in a small wound, this effect was similar across treatments. 72 hours after injection of the donor plants, plants were harvested by separating them into above-ground (leaves) and belowground (roots) portions. Root subsamples were set aside for microscopic assessment of root colonization by the fungus, and then all samples were dried at 60 °C for 72 hours, weighed, and ground into a fine powder for dye content analysis. On the day of harvest, a sample of the growth medium was collected adjacent to each plant and plated on PDA to verify that no unexpected microbes were present in the growth media.

Experimental validation

To further evaluate the integrity of the experimental treatments, we conducted a second round of experiments ("Follow-Up") at a separate time from the primary experiment ("Main"). The Follow-Up included additional treatments designed to test for potential confounding factors, such as substituting *Pleurotus ostreatus* Jacquin (Kummer)(a fungus not known to associate with live plants) for the DSE species and removing the barrier between chambers (Supplementary Table 1). While the results of the Follow-Up support the findings of the Main experiment (Supplementary Fig. 5; Supplementary Fig. 6; Supplementary Fig. 7), the two rounds were not treated as experimental replicates due to differences in implementation of the two. Specifically, fertilizer was added partway through the Follow-Up experiment (but not in the Main), and the Follow-Up included additional treatments. For these reasons, we present the Follow-Up as complementary validation rather than as a direct replicate.

Fungal colonization validation

To assess fungal colonization, we examined a subset of roots under a light microscope at 40× magnification. Initial observations were recorded as presence/absence; however, imaging was hindered by residual acid fuchsin dye. To improve visualization, roots were cleared in 2.5% potassium hydroxide (KOH) for six days and subsequently stained with Trypan Blue. After destaining, root samples were mounted on slides and scanned again at 40× magnification. Due to limitations in microscope resolution beyond this magnification, image clarity remained suboptimal. Moreover, the KOH treatment appeared to remove melanin from the DSE hyphae, reducing contrast and making the post-staining images less informative than the initial, unprocessed observations.

To confirm the potential for direct DSE hyphal continuity between plants, an additional experiment was done where two axenic sorghum seeds and the DSE were grown on 1% phytagel substrate, and a dissecting microscope was used to observe the interactions between the plants and fungi at 10X-40X magnification (Supplementary Fig. 2).

Dye quantification

We assessed dye transfer to plant tissues using absorbance-based quantification. For each sample, 0.001 g (IQR = 0.0007 g) of powdered dry tissue was suspended in 20 µL of molecular-grade water, vortexed, and centrifuged (5 min at 13,000 g). One microliter of the supernatant was analyzed on a NanoDrop One spectrophotometer (Thermo Scientific) to record absorbance between 190–900 nm.

To estimate dye concentration, we created a standard curve using 49 control samples to which known dye amounts were added post-harvest. A linear regression model was fit using absorbance at 545 nm (primary absorbance peak of acid fuchsin, Supplementary Fig. 4), yielding high predictive power (adjusted $R^2 = 0.81$; $p < 0.001$; $F(1,47) = 200.1$). This model was used to estimate dye content in experimental samples, scaled by dry sample weight (μg dye per g dry plant tissue).

Because model residuals showed slight non-normality (Shapiro–Wilk $W = 0.94$, $p = 0.017$), a $\log(x + 1)$ transformation was applied to predicted values. Shoot and root tissues were modeled separately due to differing signal quality; only shoot values are reported in the main text, while root data are provided in the Supplement (Supplementary Fig. 7; Supplementary Fig. 8).

Statistics and Reproducibility

We evaluated the effects of barrier type (Permeable, Impermeable, Axenic) and chamber (Donor, Receiver) on log-transformed dye content and dry biomass using generalized linear models (GLMs) with a quasipoisson distribution and identity or log link, respectively. Unequal sample sizes across treatment groups were accounted for in the model structure, and model-based estimates (e.g., estimated marginal means) were used to provide unbiased comparisons. Models were fit separately for the Main and Follow-Up experiments because they are not being treated as replicates. Residual diagnostics included Q–Q plots, residual vs. fitted plots, Shapiro–Wilk tests, and Breusch–Pagan tests. Estimated marginal means (EMMs) and Tukey-adjusted pairwise contrasts were calculated using the *emmeans* package. EMMs were back-transformed for visualization (μg dye or g biomass per plant). Sample sizes are shown on plots; full results are provided in the Supplementary (Supplementary Table 3; Supplementary Table 4). Each experimental unit consisted of one box containing a donor plant and a receiver plant (Supplementary Fig. 3). Replicates were defined as experimental boxes assigned to the same treatment at the same time. In the Main experiment, which was used for statistical analysis, we included three replicates each for the Permeable and Axenic treatments, and four replicates for the Impermeable treatment (Figs. 4, 5).

We conducted all data analyses in R version 4.3.0 with RStudio version 2023.06.0⁴¹. To read in the spectral data, we used the *nanodrop* package developed for this project⁴². Additional packages used include: *readr* for reading in .csv files⁴³, *stringr* for reading in and manipulating strings of data⁴⁴, *dplyr* and *tidyr* for data manipulation^{45,46}, *ggplot2* for plotting⁴⁷, and *EnvStats* for adding sample sizes to plots⁴⁸.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data are available on Zenodo⁴⁹. Sequencing data from the *Alternaria alternata* isolate were deposited in GenBank (Accession Number PQ284877.1) and in the Zenodo repository⁴⁹. All other data are available from the corresponding author (BMB) on reasonable request. Supplementary Information, including all data used (Supplementary Data 1), is available for this paper.

Code availability

All code used is available at https://github.com/beabock/Bock_2025_CommsBio and in the Zenodo repository⁴⁹.

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References

- Karst, J., Jones, M. D. & Hoeksema, J. D. Positive citation bias and overinterpreted results lead to misinformation on common

mycorrhizal networks in forests. *Nat. Ecol. Evol.* (2023) <https://doi.org/10.1038/s41559-023-01986-1>.

- Irwin, A. The ‘Mother Tree’ idea is everywhere — but how much of it is real?. *Nature* **627**, 718–721 (2024).
- Simard, S. W. et al. Net transfer of carbon between ectomycorrhizal tree species in the field. *Nature* **388**, 579–582 (1997).
- Newman, E. I. Mycorrhizal links between plants: their functioning and ecological significance. *Adv. Ecol. Res.* **18**, 243–270 (1988).
- Gilbert, L. & Johnson, D. Plant–plant communication through common mycorrhizal networks. in *Advances in Botanical Research* vol. 82 83–97 (Elsevier, 2017).
- Song, Y. Y. et al. Interplant communication of tomato plants through underground common mycorrhizal networks. *PLoS ONE* **5**, e13324 (2010).
- Henriksson, N. et al. Re-examining the evidence for the mother tree hypothesis – resource sharing among trees via ectomycorrhizal networks. *N. Phytol.* **239**, 19–28 (2023).
- Robinson, D. Mother trees, altruistic fungi, and the perils of plant personification. *Trends Plant Sci.* (2023) <https://doi.org/10.1016/j.tplants.2023.08.010>.
- Rillig, M. C., Lehmann, A., Lanfranco, L., Caruso, T. & Johnson, D. Clarifying the definition of common mycorrhizal networks. *Funct. Ecol.* 1365–2435.14545 (2024) <https://doi.org/10.1111/1365-2435.14545>.
- Rillig, M. C., Lehmann, A., Mounts, I. R. & Bock, B. M. Concurrent common fungal networks formed by different guilds of fungi. *New Phytol.* nph.20418 (2025) <https://doi.org/10.1111/nph.20418>.
- Brundrett, M. C. & Tedersoo, L. Evolutionary history of mycorrhizal symbioses and global host plant diversity. *N. Phytol.* **220**, 1108–1115 (2018).
- Smith, S. E. & Read, D. J. *Mycorrhizal Symbiosis*. (Academic Press, Amsterdam Boston, 2008).
- Orchard, S. et al. Fine endophytes (*Glomus tenue*) are related to *Mucoromycotina*, not *Glomeromycota*. *N. Phytol.* **213**, 481–486 (2017).
- Hoysted, G. A. et al. Direct nitrogen, phosphorus and carbon exchanges between *Mucoromycotina* ‘fine root endophyte’ fungi and a flowering plant in novel monoxenic cultures. *N. Phytol.* **238**, 70–79 (2023).
- Jumpponen, A. & Trappe, J. M. Dark septate endophytes: a review of facultative biotrophic root-colonizing fungi. *N. Phytol.* **140**, 295–310 (1998).
- Ruotsalainen, A. L. et al. Dark septate endophytes: mutualism from by-products?. *Trends Plant Sci.* **27**, 247–254 (2022).
- Netherway, T. et al. Pervasive associations between dark septate endophytic fungi with tree root and soil microbiomes across Europe. *Nat. Commun.* **15**, 159 (2024).
- Jumpponen, A. Dark septate endophytes - are they mycorrhizal?. *Mycorrhiza* **11**, 207–211 (2001).
- Johnson, N. C., Graham, J. H. & Smith, F. A. Functioning of mycorrhizal associations along the mutualism-parasitism continuum. *N. Phytol.* **135**, 575–585 (1997).
- Newsham, K. K. A meta-analysis of plant responses to dark septate root endophytes. *N. Phytol.* **190**, 783–793 (2011).
- Berthelot, C., Chalot, M., Leyval, C. & Blaudez, D. From Darkness to Light: Emergence of the Mysterious Dark Septate Endophytes in Plant Growth Promotion and Stress Alleviation. in *Endophytes for a Growing World* (eds. Hodkinson, T. R., Doohan, F. M., Saunders, M. J. & Murphy, B. R.) 143–164 (Cambridge University Press, 2019).
- Hawkins, H.-J. et al. Mycorrhizal mycelium as a global carbon pool. *Curr. Biol.* **33**, R560–R573 (2023).
- Klein, T., Rog, I., Livne-Luzon, S., Van Der Heijden, M. G. A. & Körner, C. Belowground carbon transfer across mycorrhizal networks among trees: Facts, not fantasy. *Open Res. Eur.* **3**, 168 (2023).

24. Simard, S. W., Ryan, T., Sm'hayetsk, L. & Perry, D. A. Opinion: Response to questions about common mycorrhizal networks. *Front. Glob. Change* **7**, 1512518 (2025).
25. Card, S. D. et al. Beneficial endophytic microorganisms of Brassica – A review. *Biol. Control* **90**, 102–112 (2015).
26. DeMers, M. *Alternaria alternata* as endophyte and pathogen. *Microbiology* **168**, 001153 (2022).
27. Thomma, B. P. H. J. *Alternaria* spp.: from general saprophyte to specific parasite: *Alternaria*. *Mol. Plant Pathol.* **4**, 225–236 (2003).
28. Oburger, E., Schmidt, H. & Staudinger, C. Harnessing belowground processes for sustainable intensification of agricultural systems. *Plant Soil* (2022) <https://doi.org/10.1007/s11104-022-05508-z>.
29. Sietiö, O. et al. Ericoid plant species and *Pinus sylvestris* shape fungal communities in their roots and surrounding soil. *N. Phytol.* **218**, 738–751 (2018).
30. Usuki, F. & Narisawa, K. A mutualistic symbiosis between a dark septate endophytic fungus, *Heteroconium chaetospira*, and a nonmycorrhizal plant, Chinese cabbage. *Mycologia* **99**, 175–184 (2007).
31. Yakti, W., Kovács, G. M., Vági, P. & Franken, P. Impact of dark septate endophytes on tomato growth and nutrient uptake. *Plant Ecol. Divers.* **11**, 637–648 (2018).
32. Carvajal Janke, N. & Coe, K. K. Evidence for a fungal loop in shrublands. *J. Ecol.* **109**, 1842–1857 (2021).
33. Rudgers, J. A. et al. Are fungal networks key to dryland primary production?. *Am. J. Bot.* **105**, 1783–1787 (2018).
34. Collins, S. L. et al. A multiscale, hierarchical model of pulse dynamics in arid-land ecosystems. *Annu. Rev. Ecol. Evol. Syst.* **45**, 397–419 (2014).
35. Collins, S. L. et al. Pulse dynamics and microbial processes in aridland ecosystems: Pulse dynamics in aridland soils. *J. Ecol.* **96**, 413–420 (2008).
36. Warren, J. M., Brooks, J. R., Meinzer, F. C. & Eberhart, J. L. Hydraulic redistribution of water from *Pinus ponderosa* trees to seedlings: evidence for an ectomycorrhizal pathway. *N. Phytol.* **178**, 382–394 (2008).
37. Hoagland, D. R. *The Water-Culture Method for Growing Plants without Soil*. vol. C347 rev 1950 (College of Agriculture, University of California, Berkeley, California, 1950).
38. Hoagland, D. R. Optimum nutrient solutions for plants. *Science* **52**, 562–564 (1920).
39. Lamit, L. J. et al. Tree genotype and genetically based growth traits structure twig endophyte communities. *Am. J. Bot.* **101**, 467–478 (2014).
40. Bock, B., Curry, L. & Gehring, C. Better utilization of inorganic nitrogen compared to organic nitrogen by a plant symbiotic fungal isolate of *Alternaria alternata*. *MicroPublication Biol.* (2025) <https://doi.org/10.17912/micropub.biology.001403>.
41. R. Core Team. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing (2023).
42. Bock, B. beabock/nanodRop. Zenodo <https://doi.org/10.5281/zenodo.12519996> (2024).
43. Wickham, H., Hester, J. & Bryan, J. readr: Read Rectangular Text Data. 2.1.5 <https://doi.org/10.32614/CRAN.package.readr> (2015).
44. Wickham, H. stringr: Simple, Consistent Wrappers for Common String Operations. 1.5.1 <https://doi.org/10.32614/CRAN.package.stringr> (2009).
45. Wickham, H., François, R., Henry, L., Müller, K. & Vaughan, D. dplyr: A Grammar of Data Manipulation. 1.1.4 <https://doi.org/10.32614/CRAN.package.dplyr> (2014).
46. Wickham, H., Vaughan, D. & Girlich, M. tidyr: Tidy Messy Data. 1.3.1 <https://doi.org/10.32614/CRAN.package.tidyr> (2014).
47. Wickham, H. *Ggplot2: Elegant Graphics for Data Analysis*. (Springer International Publishing, Cham, 2016).
48. Millard, S. P. EnvStats: Package for Environmental Statistics, Including US EPA Guidance. 3.0.0 <https://doi.org/10.32614/CRAN.package.EnvStats> (2013).
49. Bock, B. Evidence for common fungal networks among plants formed by a dark septate endophyte in *Sorghum bicolor*. Zenodo <https://doi.org/10.5281/ZENODO.15644403> (2025).

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Author contributions

C.A.G., N.C.J. and B.M.B. conceived of the idea for this project. B.M.B. led experimental design, led experimental execution and wrote the first draft of this manuscript with input from C.A.G., N.C.J. and J.D.H.

Competing interests

The authors declare no competing interests.

Additional information

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